Multiple Regression

Homework to Environmental Statistics 1

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This weeks exercises will practice the content of the lecture and tutorials and extend them in different ways:

- plotting: 3D, value-dependent symbol sizes and colours;
- interaction with quadratic terms.

Do not expect to be able to solve these exercises based on the lecture and tutorial alone! As always, the literature provided and various internet resources are your friend.

1 Exercise 1 (10 points): Seed germination

A Biologist analyzed an experiment to determine the effect of moisture content on seed germination. Eight boxes of 100 seeds each were treated with the same moisture level. 4 boxes were covered and 4 left uncovered. The process was repeated at 6 different moisture levels (nonlinear scale). Data are in **faraway**'s seeds.

1. Plot the data (2 point), using different symbols or colours for covered levels and a legend, and with properly labelled axes.
2. Analyse the data for a statistical interaction (3 points).
3. Add regression lines including 95% confidence interval to the plot (3 point).
4. Interpret your findings, using both figure and analysis (2 points).

In [14]:
```
# needed libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.anova import anova_lm
from statsmodels.formula.api import ols
import matplotlib.pyplot as plt
```

```
/Users/phoebewells/Documents/environmental_statistics/week1/data
['insulgas.csv', 'seeds.csv', 'README.md', 'sat.csv']
['insulgas.csv', 'seeds.csv', 'README.md', 'sat.csv']
```

In [15]:
```
# load seed data
seed_data = pd.read_csv("seeds.csv")
```

```
seed_data.head()
```

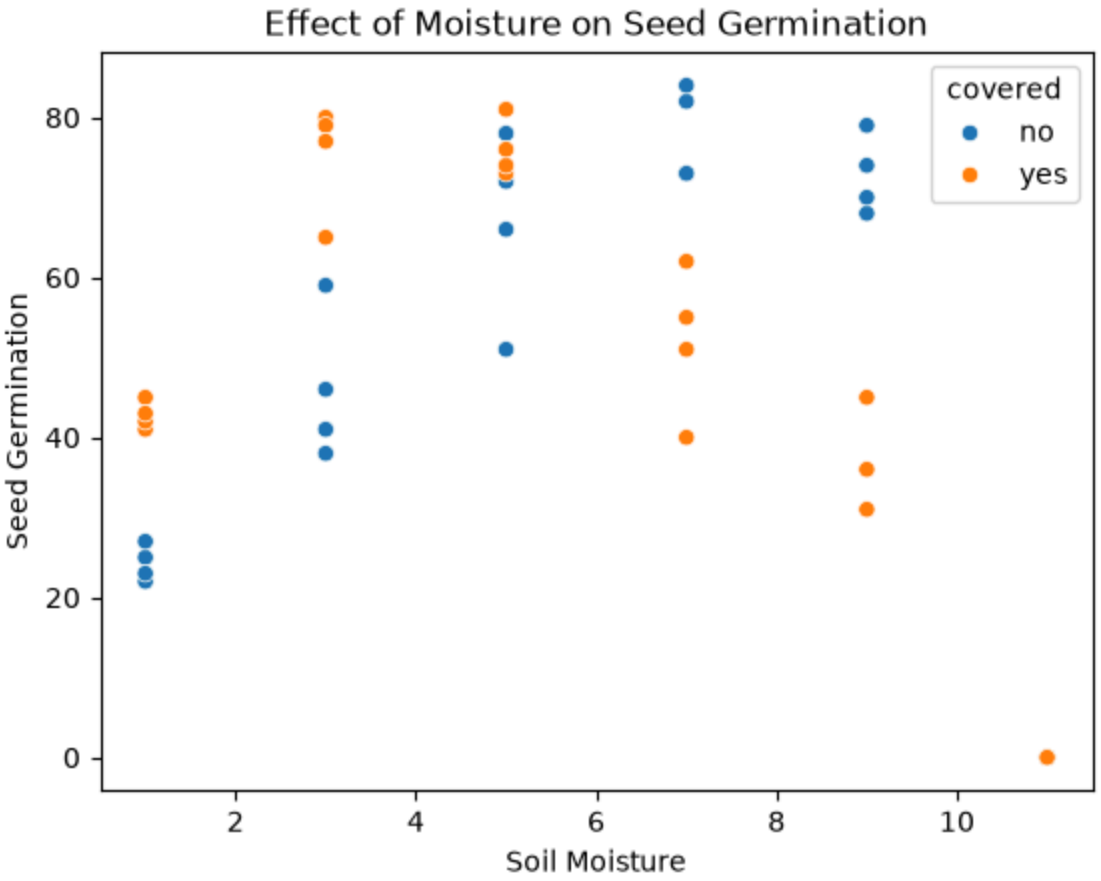
Out[15]:

	germ	moisture	covered
0	22.0	1	no
1	41.0	3	no
2	66.0	5	no
3	82.0	7	no
4	79.0	9	no

In [ ]:
```
#describe data
seed_data.describe()
```

1.1 Plot the data

In [16]:
```
# Create a scatter plot
sns.scatterplot(data = seed_data, y= 'germ', x='moisture', hue="covered")
plt.title('Effect of Moisture on Seed Germination')
plt.xlabel('Soil Moisture')
plt.ylabel('Seed Germination')
plt.show()
```



1.2 Analysis

In [18]:
```
# model with interaction
model_int = ols('germ ~ moisture *covered', data = seed_data).fit()
print(model_int.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   germ   R-squared:                       0.1
95
Model:                            OLS   Adj. R-squared:                  0.1
38
Method:                 Least Squares   F-statistic:                     3.4
63
Date:                Mon, 26 Oct 2026   Prob (F-statistic):             0.02
43
Time:                        18:29:58   Log-Likelihood:                -217.
79
No. Observations:                  47   AIC:                             44
3.6
Df Residuals:                      43   BIC:                             45
1.0
Df Model:                           3
Covariance Type:            nonrobust
================================================================================
===============
                               coef    std err          t      P>|t|      [0.
025      0.975]
--------------------------------------------------------------------------------
---------------
Intercept                   50.5083     10.740      4.703      0.000      28.
849      72.168
covered[T.yes]              26.9689     15.201      1.774      0.083      -3.
687      57.625
moisture                    -0.4250      1.556     -0.273      0.786      -3.
562       2.712
moisture:covered[T.yes]     -4.6563      2.219     -2.098      0.042      -9.
131      -0.181
==============================================================================
==
Omnibus:                       11.438   Durbin-Watson:                   1.4
68
Prob(Omnibus):                  0.003   Jarque-Bera (JB):                3.8
62
Skew:                          -0.377   Prob(JB):                        0.1
45
Kurtosis:                       1.815   Cond. No.                         3
6.2
==============================================================================
==

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is corre
ctly specified.
```

1.3 Regression plot

In [20]:
```
# plot the regression line on the data including 95% confidence interval

fig, ax = plt.subplots(figsize=(6, 6))
```

```
colors = {'no':'tab:blue', 'yes':'tab:orange'}
ax.scatter(seed_data['moisture'], seed_data['germ'], c=seed_data['covered'].

#min and max of uncovered seeds
min1 = min(seed_data.moisture[seed_data.covered=="no"])
max1 = max(seed_data.moisture[seed_data.covered=="no"])
#generate points between min and max
new_moisture_no = np.linspace(min1, max1, num=101) # numpy.ndarray

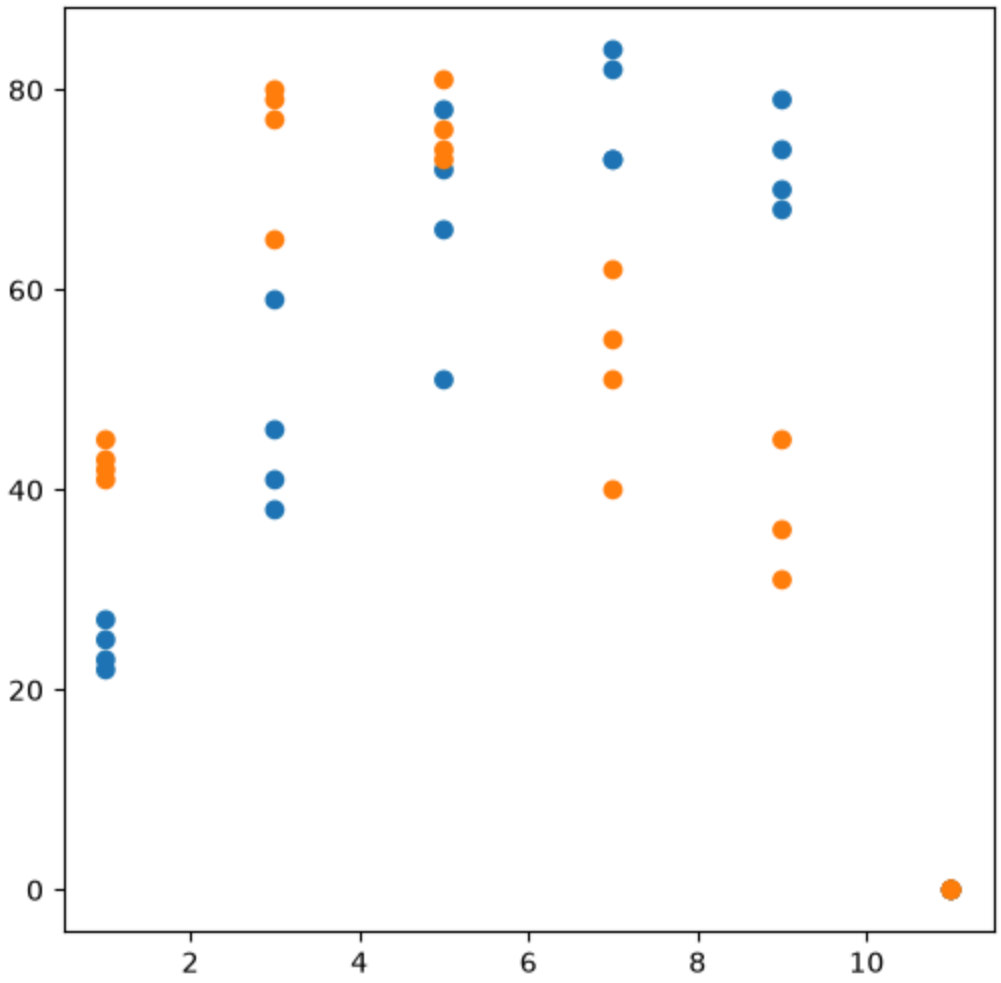
#min and max of covered seeds
min2 = min(seed_data.moisture[seed_data.covered=="yes"])
max2 = max(seed_data.moisture[seed_data.covered=="yes"])
#generate points between min and max
new_moisture_yes = np.linspace(min2, max2, num=101) # numpy.ndarray

# Convert to a dataframe
new_moisture_no_df = pd.DataFrame(list(new_moisture_no), columns=["moisture"
new_moisture_no_df['covered'] = "no"
new_moisture_yes_df = pd.DataFrame(list(new_moisture_yes), columns=["moistur
new_moisture_yes_df['covered'] = "yes"

# Prediction on generated points
preds_no = model_int.get_prediction(new_moisture_no_df).summary_frame(alpha=
preds_yes = model_int.get_prediction(new_moisture_yes_df).summary_frame(alph
preds_yes.head()
```

Out[20]:

	mean	mean_se	mean_ci_lower	mean_ci_upper	obs_ci_lower	obs_ci_upper
0	72.395887	9.425306	53.387946	91.403828	16.564769	128.227005
1	71.887757	9.296386	53.139809	90.635705	16.144619	127.630896
2	71.379627	9.168384	52.889819	89.869435	15.722778	127.036477
3	70.871497	9.041339	52.637900	89.105095	15.299238	126.443757
4	70.363368	8.915293	52.383967	88.342769	14.873992	125.852743



In [21]:
```
# main plot
fig, ax = plt.subplots(figsize=(6, 6))
colors = {'no':'tab:blue', 'yes':'tab:orange'}
ax.scatter(seed_data['moisture'], seed_data['germ'], c=seed_data['covered'].

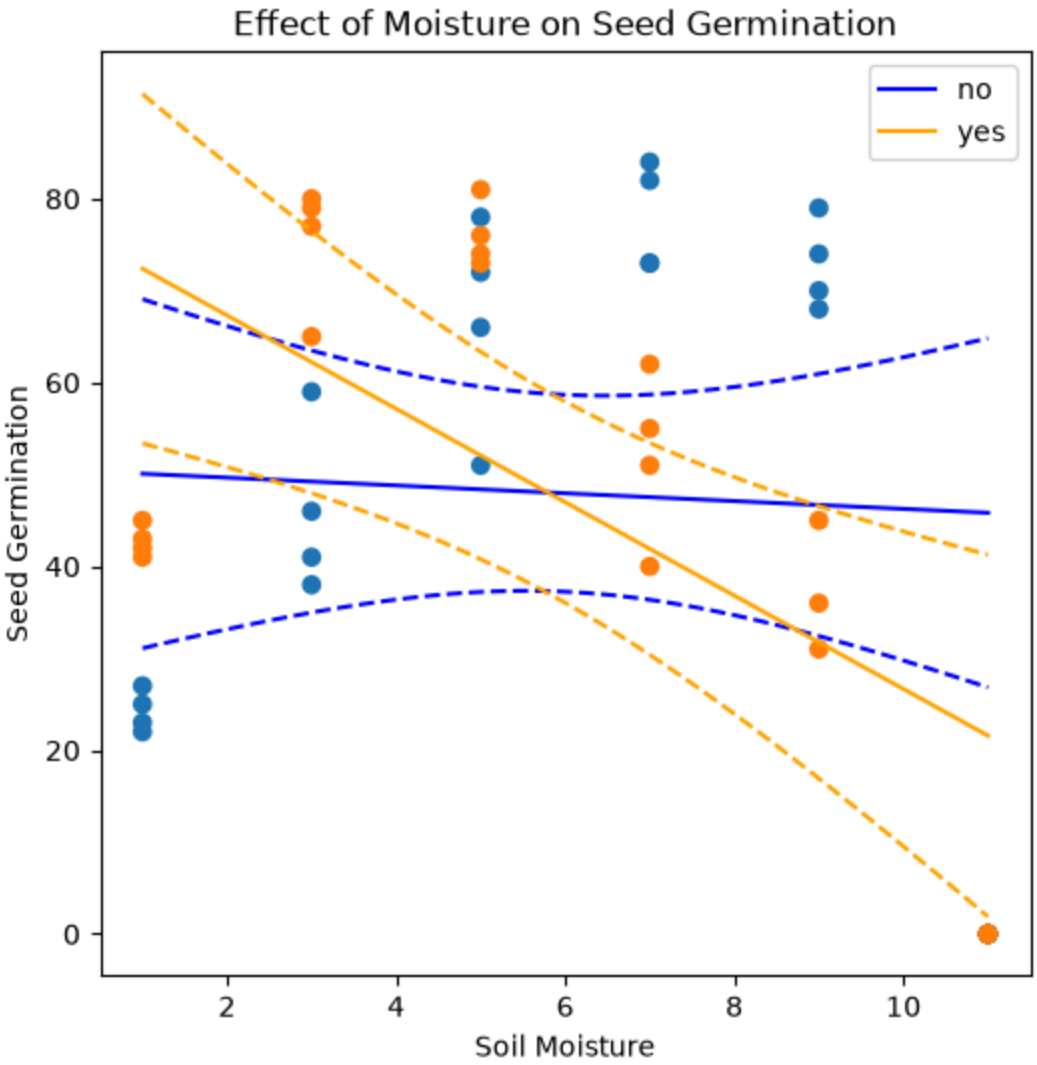
# Before
plt.plot(new_moisture_no_df['moisture'], preds_no['mean'], color='blue', lab
plt.plot(new_moisture_no_df['moisture'], preds_no['mean_ci_lower'], color='b
plt.plot(new_moisture_no_df['moisture'], preds_no['mean_ci_upper'], color='b

# After
plt.plot(new_moisture_yes_df['moisture'], preds_yes['mean'], color='orange',
plt.plot(new_moisture_yes_df['moisture'], preds_yes['mean_ci_lower'], color=
plt.plot(new_moisture_yes_df['moisture'], preds_yes['mean_ci_upper'], color=

# Set common labels
ax.set_ylabel('Germination')
ax.set_xlabel('moisture')
leg = plt.legend()

plt.title('Effect of Moisture on Seed Germination')
plt.xlabel('Soil Moisture')
plt.ylabel('Seed Germination')

plt.show()
```



1.4 Interpret

Upon initial observation of the observed points, it appears that there might be some correlation between moisture level, covering, and germination. However, the graph does not appear to show any strong correlation. The initial summary of the observed data yields significant p values only for the interaction between moisture and covering, implying that the effect of moisture on germination is dependent on whether or not the seeds are covered. The final graph displaying the regression lines of the predicted values confirms this, as the germination of the covered seeds appear negatively correlated whereas the germination for the noncovered stay relatively the same across moisture levels.

2 Exercise 2: SAT scores (14 points)

Public schools in the USA (as everywhere else) are under scrutiny for their performance: do pupils get better grades when the teachers are better paid? Using 1994/1995 data,

we shall compare the 50 states in the total performance of pupils (the SAT score `total`) as a function of proportion of eligible pupils (`takers`) and teacher's average annual `salary`.

1. Plot the data, using colour *and* symbol size to indicate SAT scores and putting the two predictors onto x- and y- axis. This imitates a 3D-plot! (Hint: For the colour option you may want to re-scale the response values to between 0.2 and 1. That is similar to rescaling to [0,1], times the new intended range (i.e. 0.8) plus the offset (0.2). This is also about learning how to find help online.)(Hint 2: For colour, you can use `cmap='viridis'`) (**4 points**)

2. Based on the plot, can you make any statement about the effect of takers and salary on SAT-scores? (**2 points**)

3. Analyse total SAT-score for effects of proportion of takers and salary, and their interaction. Describe, briefly, what you find. As always, do **not** use the names of the variables in your answer, i.e. no "there is an effect of `takers`", but "there is an effect of the number of pupils taking the exam"! (And be more specific than that, of course.) (**4 points**)

4. Make a "3D"-plot of the model with interaction, i.e. either an actual 3D-plot, or a "topographic map"-like plot (with contours). Indiate where the data are (e.g. as little crosses). Re-examing your interpretation from the previous step and formulate what the interaction of the two predictors means to pupil's SAT-scores. (Hint: Do not worry about prediction error. It is difficult to visualise that in 3D.) (**4 points**)

2.1 Plotting

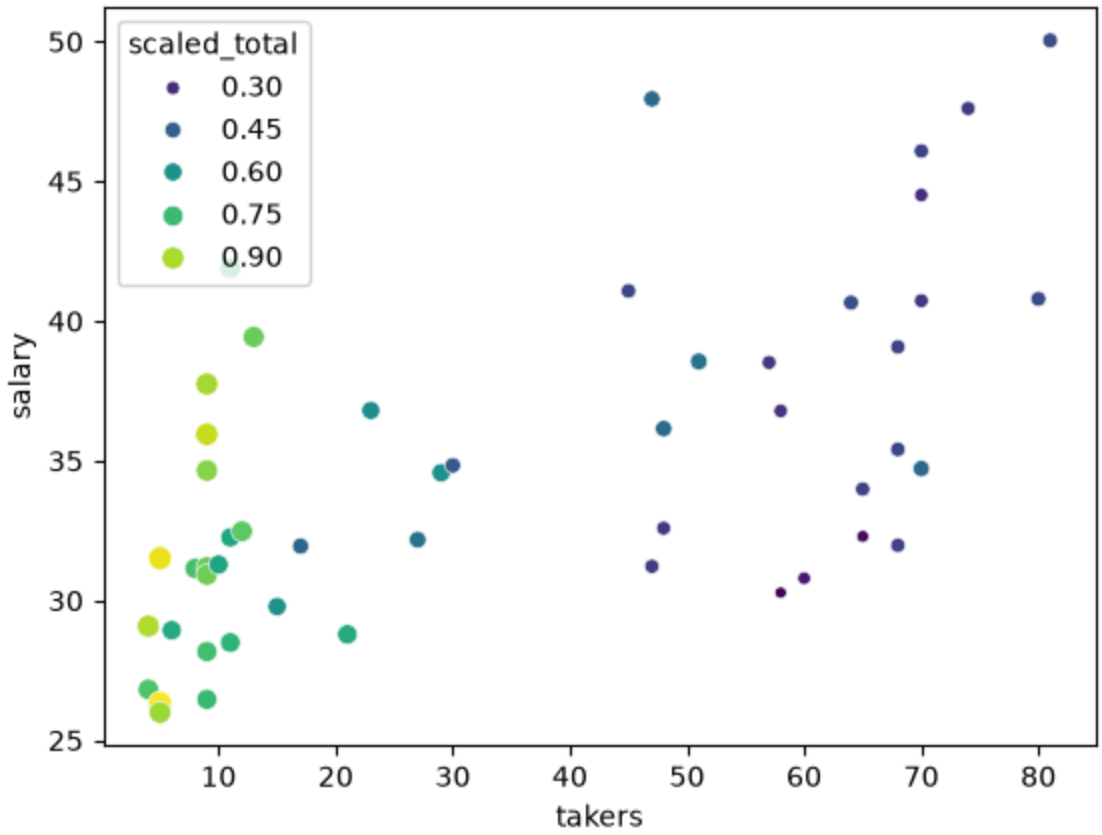
In [7]:
```
# load data nad make plot
sat_data = pd.read_csv("sat.csv")
display(sat_data.head())

total_min = sat_data.total.min()
total_max = sat_data.total.max()

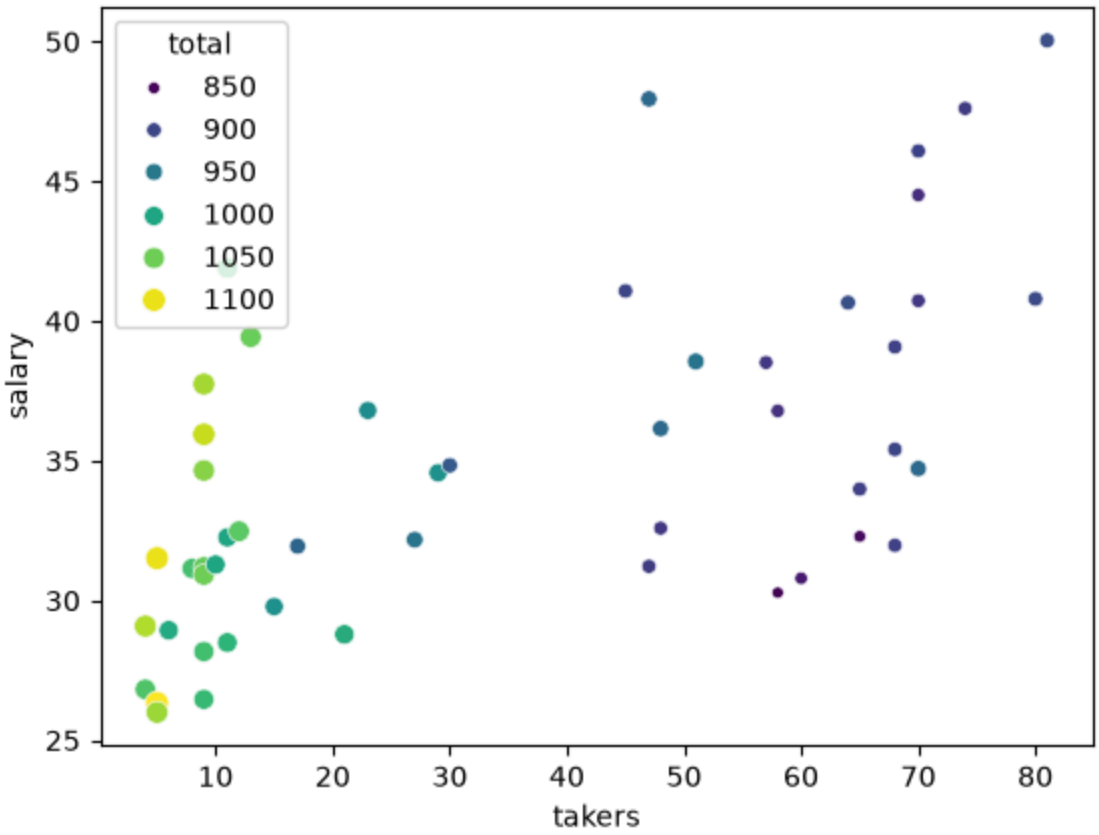
sat_data["scaled_total"] = (0.2 + (sat_data.total - total_min)/(total_max -

sns.scatterplot(x = "takers", y = "salary", hue = "scaled_total", size = "sc
plt.show()
```

	state	expend	ratio	salary	takers	verbal	math	total
0	Alabama	4.405	17.2	31.144	8	491	538	1029
1	Alaska	8.963	17.6	47.951	47	445	489	934
2	Arizona	4.778	19.3	32.175	27	448	496	944
3	Arkansas	4.459	17.1	28.934	6	482	523	1005
4	California	4.992	24.0	41.078	45	417	485	902



Out[7]: <Axes: xlabel='takers', ylabel='salary'>



2.2 Visual interpretation

The plot appears to show a negative correlation between the proportion of takers and the total SAT scores and very weak correlation between takers and total SAT score.

2.3 Analysis

In [11]:
```
sat_int = ols('total ~ takers*salary', data = sat_data).fit()
print(sat_int.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  total   R-squared:                       0.823
Model:                            OLS   Adj. R-squared:                  0.811
Method:                 Least Squares   F-statistic:                     71.14
Date:                Sun, 25 Oct 2026   Prob (F-statistic):           2.64e-17
Time:                        16:02:18   Log-Likelihood:                -242.95
No. Observations:                  50   AIC:                             493.9
Df Residuals:                      46   BIC:                             501.6
Df Model:                           3
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
Intercept      1082.3456     54.352     19.914      0.000     972.941    1191.750
takers           -5.0268      1.089     -4.616      0.000      -7.219      -2.835
salary           -0.7200      1.697     -0.424      0.673      -4.136       2.696
takers:salary     0.0648      0.031      2.108      0.041       0.003       0.127
==============================================================================
Omnibus:                        0.109   Durbin-Watson:                   2.348
Prob(Omnibus):                  0.947   Jarque-Bera (JB):                0.226
Skew:                           0.101   Prob(JB):                        0.893
Kurtosis:                       2.740   Cond. No.                     2.04e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.04e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The ols model with interaction shows that the proportion of eligible pupils who take the exam has significant negative correlation with the total SAT score. Teacher salary on its

own shows no significant correlation with total SAT score. There is significant interaction between the two variables, meaning that the effect of teacher salary on total SAT score is dependent on the proportion of eligible takers and vice versa.

2.4 3D-plotting of fitted model

In [43]:
```
# Either as a 3D surface plot or as contour plot
fig = plt.figure(figsize=(15, 8))
ax = fig.add_subplot(111, projection = "3d")

scatter = ax.scatter(sat_data["takers"], sat_data["salary"], sat_data["scale
ax.set_zlabel("Total SAT score")
ax.set_xlabel("Takers (%)")
ax.set_ylabel("Salary ($1000)")
fig.colorbar(scatter, ax=ax, label="Total SAT score")
ax.set_title("Effect of Teachers' Salary and Test Participation on SAT Score

plt.show()
```

